

EAS 501 Lecture Notes 09/14/2026

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Outline

- Chapter 14. Vectors in Three-dimension

14.2 Dot and cross products, 14.3 Cartesian coordinates

For $\mathbf{u} = (x_1, y_1, z_1)^T$, $\mathbf{v} = (x_2, y_2, z_2)^T$, the dot product is defined as

$$\text{Geometric definition : } \mathbf{u} \cdot \mathbf{v} := |\mathbf{u}| \cdot |\mathbf{v}| \cdot \cos \theta, \quad (1)$$

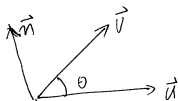
$$\text{Analytic definition : } \mathbf{u} \cdot \mathbf{v} := x_1x_2 + y_1y_2 + z_1z_2. \quad (2)$$

The cross product is defined as

$$\text{Geometric definition : } \mathbf{u} \times \mathbf{v} := (|\mathbf{u}| \cdot |\mathbf{v}| \cdot \sin \theta) \mathbf{n}, \quad (3)$$

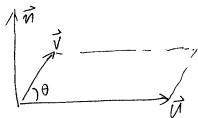
$$\text{Analytic definition : } \mathbf{u} \times \mathbf{v} := \begin{pmatrix} y_1z_2 - y_2z_1 \\ x_2z_1 - x_1z_2 \\ x_1y_2 - x_2y_1 \end{pmatrix}, \quad (4)$$

in which $\mathbf{n}$ is the unit normal vector, with the orientation given by the right hand rule.

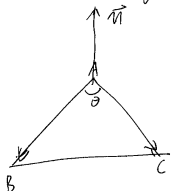


$$\vec{u} \cdot \vec{v} = |\vec{u}| \cdot |\vec{v}| \cdot \cos \theta$$

$$|\vec{u} \times \vec{v}| = |\vec{u}| \cdot |\vec{v}| \cdot \sin \theta$$



$$\text{Area} = |\vec{n}| = |\vec{u} \times \vec{v}| = |\vec{u}| \cdot |\vec{v}| \cdot \sin \theta$$



$$\vec{n} = \vec{AB} \times \vec{AC},$$

$$S_{\triangle ABC} = \frac{1}{2} |\vec{AB}| \cdot |\vec{AC}| \cdot \sin \theta$$

$$= \frac{1}{2} |\vec{n}|$$

$$= \frac{1}{2} |\vec{AB} \times \vec{AC}|,$$

In more details, the analytic derivation for the cross product is given by

$$\begin{aligned}
 \mathbf{u} \times \mathbf{v} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \end{vmatrix} \\
 &= \mathbf{i} \cdot \begin{vmatrix} y_1 & z_1 \\ y_2 & z_2 \end{vmatrix} - \mathbf{j} \cdot \begin{vmatrix} x_1 & z_1 \\ x_2 & z_2 \end{vmatrix} + \mathbf{k} \cdot \begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix} \\
 &= \begin{pmatrix} y_1 z_2 - y_2 z_1 \\ x_2 z_1 - x_1 z_2 \\ x_1 y_2 - x_2 y_1 \end{pmatrix}, \tag{5}
 \end{aligned}$$

in which $\mathbf{i} = (1, 0, 0)^T$, $\mathbf{j} = (0, 1, 0)^T$, $\mathbf{k} = (0, 0, 1)^T$ are the three fundamental vectors in R^3 .

Example 14.3 # 4 (b)

Find the area of triangle $\triangle ABC$, with $A(2, -2, 1)$, $B(4, 0, 3)$, $C(2, 3, 5)$.

Solution. We have $\vec{AB} = (2, 2, 2)^T$, $\vec{AC} = (0, 5, 4)^T$, so that

$$\begin{aligned} \mathbf{n} &= \vec{AB} \times \vec{AC} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 2 & 2 \\ 0 & 5 & 4 \end{vmatrix} \\ &= \mathbf{i} \cdot \begin{vmatrix} 2 & 2 \\ 5 & 4 \end{vmatrix} - \mathbf{j} \cdot \begin{vmatrix} 2 & 2 \\ 0 & 4 \end{vmatrix} + \mathbf{k} \cdot \begin{vmatrix} 2 & 2 \\ 0 & 5 \end{vmatrix} = \begin{pmatrix} -2 \\ -8 \\ 10 \end{pmatrix}. \end{aligned}$$

Example 14.3 # 4 (b), continued

In turn, we arrive at

$$\begin{aligned} S_{\Delta ABC} &= \frac{1}{2} |\vec{AB}| \cdot |\vec{AC}| \cdot \sin A = \frac{1}{2} |\mathbf{n}| \\ &= \frac{1}{2} |(-2, -8, 10)^T| \\ &= \frac{1}{2} \sqrt{(-2)^2 + (-8)^2 + 10^2} \\ &= \frac{1}{2} \sqrt{168} = \frac{1}{2} \cdot 2\sqrt{42} = \sqrt{42}. \end{aligned}$$

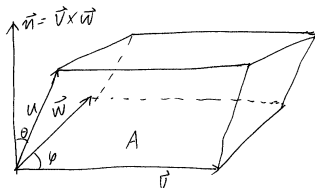
14.4 Multiple products

scalar triple product: $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$

vector triple product: $(\mathbf{u} \times \mathbf{v}) \times \mathbf{w}$

Facts:

- ▶ $(\mathbf{u} \times \mathbf{v}) \times \mathbf{w} \neq \mathbf{u} \times (\mathbf{v} \times \mathbf{w})$
- ▶ $\mathbf{u} \times (\mathbf{v} \times \mathbf{w}) = (\mathbf{u} \cdot \mathbf{w})\mathbf{v} - (\mathbf{u} \cdot \mathbf{v})\mathbf{w}$



$$A = |\vec{v}| \cdot |\vec{w}| \cdot \sin \phi = |\vec{v} \times \vec{w}|$$

$$V = A \cdot |\vec{u}| \cdot \cos \theta = |\vec{u} \cdot (\vec{v} \times \vec{w})|$$

the volume of parallelepiped.

Example 14.4 #1 (b)

Consider $\mathbf{u} = \mathbf{i} - \mathbf{j} = (1, -1, 0)^T$, $\mathbf{v} = 2\mathbf{i} + \mathbf{j} + 3\mathbf{k} = (2, 1, 3)^T$, $\mathbf{w} = \mathbf{j} - \mathbf{k} = (0, 1, -1)^T$. Compute $\mathbf{v} \times \mathbf{w}$, $\mathbf{u} \times \mathbf{v}$, $(\mathbf{u} \times \mathbf{v}) \times \mathbf{w}$, $\mathbf{u} \times (\mathbf{v} \times \mathbf{w})$.

$$\mathbf{v} \times \mathbf{w} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 1 & 3 \\ 0 & 1 & -1 \end{vmatrix} = \begin{pmatrix} -4 \\ 2 \\ 2 \end{pmatrix}, \quad (6)$$

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -1 & 0 \\ 2 & 1 & 3 \end{vmatrix} = \begin{pmatrix} -3 \\ -3 \\ 3 \end{pmatrix}, \quad (7)$$

Example 14.4 #1 (b), continued

$$\mathbf{u} \times (\mathbf{v} \times \mathbf{w}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -1 & 0 \\ -4 & 2 & 2 \end{vmatrix} = \begin{pmatrix} -2 \\ -2 \\ -2 \end{pmatrix}, \quad (8)$$

$$(\mathbf{u} \times \mathbf{v}) \times \mathbf{w} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -3 & -3 & 3 \\ 0 & 1 & -1 \end{vmatrix} = \begin{pmatrix} 0 \\ -3 \\ -3 \end{pmatrix}. \quad (9)$$

14.5 Differentiation of a vector function

Consider a vector function $\mathbf{r}(t) = (x(t), y(t), z(t))^T$, we define

$$\text{tangent vector : } \mathbf{r}'(t) = (x'(t), y'(t), z'(t))^T, \quad (10)$$

$$\text{unit tangent vector : } \mathbf{T}(t) = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|}. \quad (11)$$

If $\mathbf{r}(t)$ is the position function, $\mathbf{v}(t) = \mathbf{r}'(t)$ is the velocity vector function, and $\mathbf{a}(t) = \mathbf{v}'(t) = \mathbf{r}''(t)$ is the acceleration vector function.

Example 14.5 # 1 (e)

Consider

$$\mathbf{u}(\tau) = e^{-\tau}(\cos(2\tau)\mathbf{i}, \sin(2\tau)\mathbf{j})^T = (e^{-\tau} \cos(2\tau), e^{-\tau} \sin(2\tau))^T.$$

$$\mathbf{u}'(\tau) = \begin{pmatrix} e^{-\tau}[-\cos(2\tau) - 2\sin(2\tau)] \\ e^{-\tau}[-\sin(2\tau) + 2\cos(2\tau)] \end{pmatrix}, \quad (12)$$

$$\mathbf{u}''(\tau) = \begin{pmatrix} e^{-\tau}[4\sin(2\tau) - 3\cos(2\tau)] \\ e^{-\tau}[-4\cos(2\tau) - 3\sin(2\tau)] \end{pmatrix}, \quad (13)$$

so that

$$\begin{aligned} |\mathbf{u}''(\tau)| &= \left(e^{-2\tau} \left([4\sin(2\tau) - 3\cos(2\tau)]^2 \right. \right. \\ &\quad \left. \left. + [-4\cos(2\tau) - 3\sin(2\tau)]^2 \right) \right)^{\frac{1}{2}} \\ &= \sqrt{25e^{-2\tau}} = 5e^{-\tau}. \end{aligned} \quad (14)$$